Kubernetes and Istio

04 - Managing Pods with Deployments

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A **Deployment** is a Kubernetes API object that allows you to declare the desired state of your application, delegating the complex mechanics of getting to that state to Kubernetes itself.
While you can run applications by creating individual Pods, managing them manually during updates or node failures is impractical. Deployments solve this by providing declarative updates, horizontal scaling, and self-healing mechanisms.
Rather than manually replacing old Pods with new ones, you simply update the Deployment configuration, and the Kubernetes Deployment controller automates the transition at a controlled rate. Because Deployments treat individual Pods as fungible (easily replaceable) replicas, they are the standard choice for running stateless workloads.

Here is a minimal YAML declaration of a Deployment:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
      - name: nginx
        image: nginx:1.21
```

Understanding Deployment `spec`

Field	Description
`replicas`	The desired number of identical Pods to run concurrently. Kubernetes actively maintains this exact count.
`selector`	A label query that tells the Deployment which Pods belong to it. This must match the labels defined in the `template`.
`template`	The blueprint (Pod Template) used to stamp out new Pods. It contains its own metadata (labels) and specifications (containers, images, volumes).
`strategy`	Defines the exact methodology Kubernetes will use to replace old Pods with new ones when the Pod template is updated.

1. Deployment scaling

Scaling a Deployment up or down is as simple as modifying the `replicas` field in the YAML manifest or executing a command like `kubectl scale deployment <name> --replicas 5`.

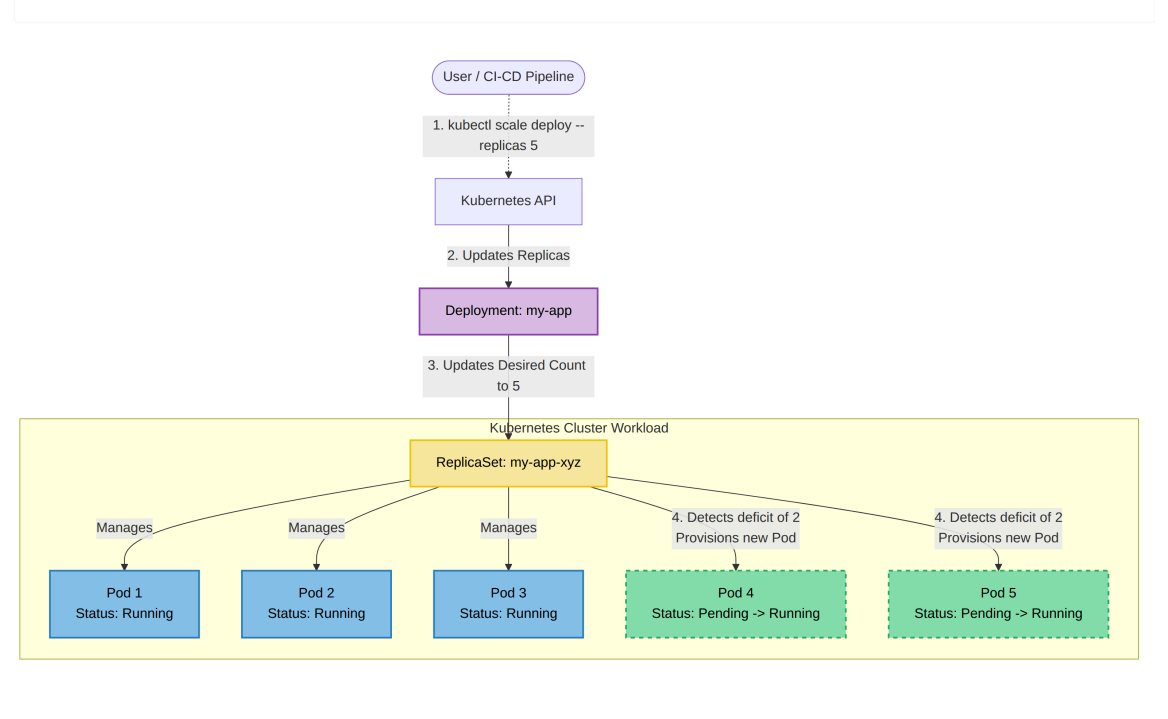
To understand how this works under the hood, it is essential to understand **ReplicaSets**. A ReplicaSet is a lower-level Kubernetes object whose sole responsibility is to ensure that a specified number of Pod replicas are running at any given time.

Deployments do not manage Pods directly. Instead, a **Deployment manages a ReplicaSet**, and that ReplicaSet directly creates and manages the Pods. When you scale a Deployment, the Deployment controller simply updates the desired replica count on the underlying ReplicaSet. The ReplicaSet controller then provisions or terminates Pods to match that new target.

Deployments can be **autoscaled** by using the following object: `HorizontalPodAutoscaler` (HPA). With HPA, you can define a controller which scales up or down your application *automagically* when a specified metric (e.g. response time) differs from its desired value.
Since this object is really complex to manage and define and its usage differs completely from application to application, it is not explained in this guide.
You can find more information about it at Kubernetes documentation - HPA.

Example:

```
kubectl scale deployment product-service --replicas 5
```


User / CI-CD Pipeline
1. kubectl scale deploy --replicas 5
Kubernetes API
2. Updates Replicas
Deployment: my-app
3. Updates Desired Count to 5
Kubernetes Cluster Workload
ReplicaSet: my-app-xyz
Manages
Manages
Manages
4. Detects deficit of 2 Provisions new Pod
4. Detects deficit of 2 Provisions new Pod
Pod 1 Status: Running
Pod 2 Status: Running
Pod 3 Status: Running
Pod 4 Status: Pending -> Running
Pod 5 Status: Pending -> Running

2. Deployment update strategy

The true power of a Deployment lies in its ability to update workloads seamlessly. When you change the Pod template (for instance, updating the container image to a new version), a rollout is triggered. Kubernetes currently supports two main native update strategies:

- `Recreate` : destroy all current active pods and start the new version
- `RollingUpdate` : gradually stop pods and start the updated version

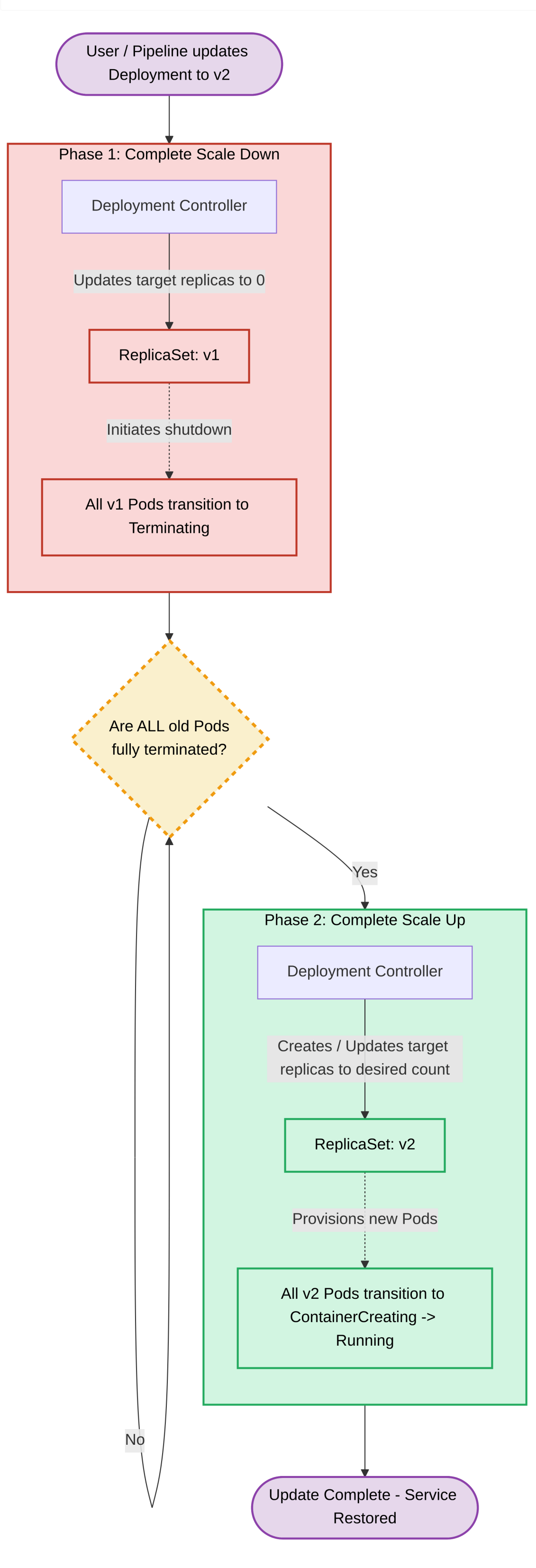
2.1 Recreate strategy

With the Recreate strategy, the Deployment controller scales the existing ReplicaSet down to zero, destroying all currently running Pods simultaneously. Only after the old Pods are completely terminated does it create a new ReplicaSet and spin up the new Pods.

- **Pros**: Prevents version collisions. The old and new versions of the application never run at the same time.
- **Cons**: Causes complete service downtime while the replacement Pods are provisioning.

YAML declaration

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-app-recreate
spec:
  replicas: 3
  strategy: # The update strategy is defined here
    type: Recreate
  selector:
    matchLabels:
      app: my-web-app
  template:
    metadata:
      labels:
        app: my-web-app
    spec:
      containers:
      - name: web-container
        image: my-app:v2
```


User / Pipeline updates Deployment to v2
Phase 1: Complete Scale Down
Deployment Controller
Updates target replicas to 0
ReplicaSet: v1
Initiates shutdown
All v1 Pods transition to Terminating
Are ALL old Pods fully terminated?
Yes
No
Phase 2: Complete Scale Up
Deployment Controller
Creates / Updates target replicas to desired count
ReplicaSet: v2
Provisions new Pods
All v2 Pods transition to ContainerCreating -> Running
Update Complete - Service Restored

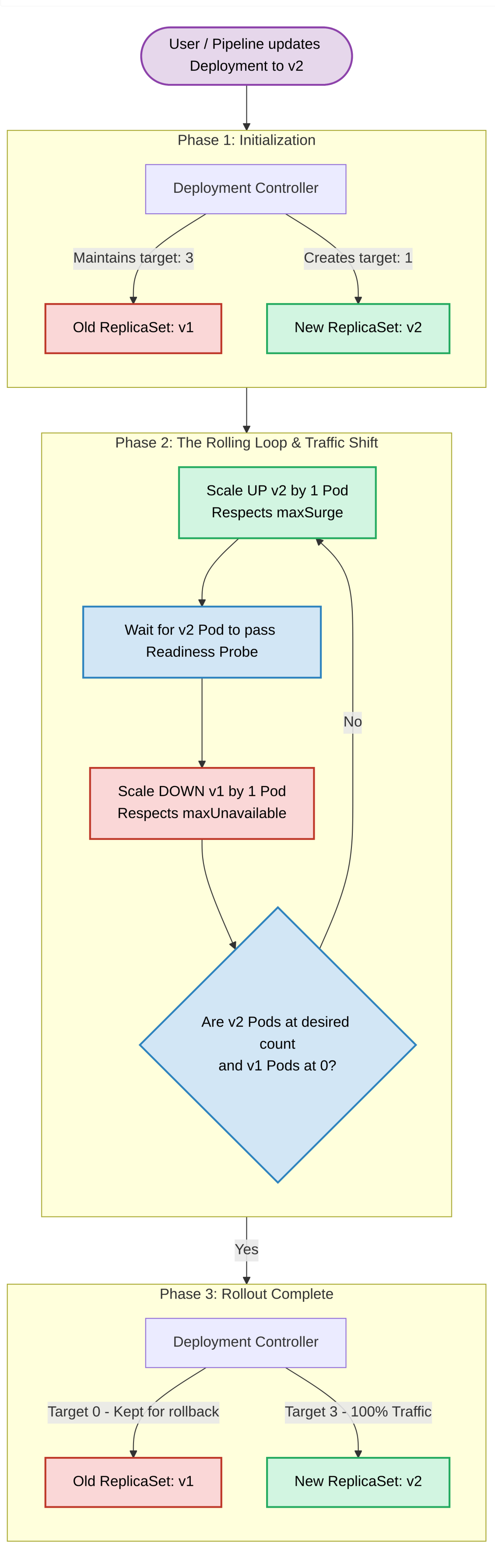
2.2 RollingUpdate strategy

The RollingUpdate strategy (which is the default) guarantees zero downtime. It gradually removes old Pods while simultaneously starting new ones. During this process, client traffic is distributed across both the old and new versions of the application. The speed and safety of this rollout are controlled by two parameters:

- **maxSurge**: The maximum number of Pods that can be created above the desired replica count during the update.
- **maxUnavailable**: The maximum number of Pods that can be unavailable during the update process.

YAML declaration:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-app-rolling
spec:
  replicas: 4
  strategy:
    type: RollingUpdate # Define the update strategy here
    rollingUpdate:
      # How many extra Pods can be created above the desired replica count
      maxSurge: 1
      # How many Pods can be missing from the desired replica count
      maxUnavailable: 25%
  selector:
    matchLabels:
      app: my-web-app
  template:
    metadata:
      labels:
        app: my-web-app
    spec:
      containers:
      - name: web-container
        image: my-app:v2
```


User / Pipeline updates Deployment to v2
Phase 1: Initialization
Deployment Controller
Maintains target: 3
Creates target: 1
Old ReplicaSet: v1
New ReplicaSet: v2
Phase 2: The Rolling Loop & Traffic Shift
Scale UP v2 by 1 Pod Respects maxSurge
Wait for v2 Pod to pass Readiness Probe
Scale DOWN v1 by 1 Pod Respects maxUnavailable
Are v2 Pods at desired count and v1 Pods at 0?
No
Yes
Phase 3: Rollout Complete
Deployment Controller
Target 0 - Kept for rollback
Target 3 - 100% Traffic
Old ReplicaSet: v1
New ReplicaSet: v2

2.3 Advanced update strategies

While Recreate and RollingUpdate are built into the Deployment object, the flexibility of Kubernetes allows you to construct more advanced rollout patterns using a combination of multiple Deployments, Services, and Ingress controllers:

Strategy	Concept
Canary	Releasing a new version to a very small subset of users to validate stability before a full rollout.
A/B testing	Routing specific user segments to a new version based on conditions like location or headers to measure performance/engagement.
Blue/Green	Deploying the entire new version alongside the old one, then instantly flipping a network switch (Service selector) to route all traffic to the new version.
Shadowing	Mirroring live production traffic to a hidden new version to test how it handles real-world loads without impacting end users.

Update rollbacks

Deployments act as a safety net. If you deploy a faulty version of an application (e.g., a version that crashes on startup or fails its readiness probes), the Deployment controller will halt the `RollingUpdate` process, preventing a complete cluster outage.

Kubernetes preserves a history of previous ReplicaSets. This allows you to quickly revert to a known, stable state using the rollback command:

```
kubectl rollout undo deployment DEPLOYMENT-NAME
```

This command immediately tells the Deployment to scale the new, faulty ReplicaSet back down to zero and scale the previous, stable ReplicaSet back up to the desired target, restoring service reliability.